

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Logistics Management)

as per NEP 2020

w.e.f session 2024-25

Course Outline for B.Voc (Logistics Management)
as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BLM 101	Core Discipline	Introduction to Logistics Management	4	-	4	80	20	-	-	100
BLM 102	Core Discipline	Principles of Management	4	-	4	80	20	-	-	100
BLM 103	Core Discipline	Computerised Accounting and Enterprise Resource Planning	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BLM 201	Core Discipline	Transportation and Distribution Management	4	-	4	80	20	-	-	100
BLM 202	Core Discipline	Inventory Management	4	-	4	80	20	-	-	100
BLM 203	Core Discipline	Warehouse Management	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Logistics Assistant

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Logistics Management = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (LOGISTICS MANAGEMENT) 2024-2025

PROGRAMME OUTCOMES

PO1. The course aims to provide comprehensive knowledge and coherent understanding of logistics sector operations and link theory with practice in the context of competency development for logistics employees.

PO2. The course further aims to provide deeper understanding of the disciplinary, multidisciplinary, value-added, skill enhancement and ability enhancement courses with practical orientation to make the students ready to make a mark in the industry with strong human and moral values.

PO3. The course will help develop skills related to different job roles such as logistics assistant, logistics operator, logistics manager including knowledge required for undertaking self- employment initiatives, and knowledge and mindset required for entrepreneurship involving enterprise creation, improved product development, or a new mode of organization.

PO4. To enhance skills for efficiently using different software, interpret and use various acts and laws related to logistics sector.

PO5. To develop communication skills and build confidence to face the challenges of the job market.

Semester I

	COURSE TITLE	INTRODUCTION TO LOGISTICS MANAGEMENT
1.	**Course Code**	BLM 101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lecture) (Practical)**	60 hours of lectures for Theory (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80 TH + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Logistics Assistant
7.	**Course Objectives**	The objective of this paper is to provide students with conceptual knowledge of the basics of logistics management, key components of logistics, customer- centric logistics strategies, and functions and services of third-party and fourth-party logistics providers.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Students will develop a comprehensive understanding of the fundamentals of logistics management, its history, objectives, and the critical elements that constitute the logistics system within retail and broader marketing contexts. CO2. Students will be able to analyze how logistics contributes to competitive advantage in business, understanding the integrated logistics management process and its role in creating efficient operations. CO3. Students will gain the ability to formulate and implement logistics strategies and will understand the strategic role of logistics in business decisions. CO4. Students will understand the reasons for and the processes involved in outsourcing logistics, including the functions and services of third-party and fourth-party logistics providers. CO5. Students will be able to design and evaluate customer service frameworks within logistics, recognizing the importance of the order cycle system and distribution channel design. CO6. Students will stay abreast of emerging concepts in logistics and will be skilled in managing and coordinating comprehensive logistics activities to achieve seamless and integrated operations.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire iv) Compulsory Question No. 1. All questions carry equal marks.

11. Course Content

UNIT – I

Logistics: Definition - History and Evolution – Objectives – Elements - activities importance - The work of logistics - Logistics interface with marketing - Retails logistics.

UNIT – II

Logistics Management: Definition - Evolution of the concept - model – process - activities. Achievement of competitive advantage through logistics Framework - Role of Logistics management - Integrated Logistics Management.

UNIT – III

Logistics Strategy: Strategic role of logistics – Definition - Role of logistics managers in strategic decisions - Strategy options, Lean strategy, Agile Strategies & Other strategies - Designing & implementing logistical strategy
- Emerging concept in logistics.

UNIT – IV

Outsourcing Logistics: Reasons - Third party logistics provider - Fourth party Logistics providers (4PL) –Stages - Role of logistics providers. Quality Customer Service & Integrated Logistics: Customer service - importance elements - the order cycle system - distribution channels - Functions performed - Types designing.

12. Suggested Resources for Reading

1. Bloomberg, D.J., LeMay, S., Logistics, Prentice-Hall of India Pvt Ltd.
2. Bowersox, D.J. & Closs, D.J., Logistical Management, Tata McGraw Hill Publishing Co. Ltd.
3. Ailawadi, S.C. & Singh, R., Logistics Management, Prentice-Hall of India Pvt. Ltd.
4. Waters, D., Logistics, Palgrave Macmillan

Note: Latest edition of text books may be used.

	COURSE TITLE	PRINCIPLES OF MANAGEMENT
1.	**Course Code**	BLM 102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. i) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Logistics Assistant
7.	**Course Objective**	The objective of this paper is to help students to understand the fundamental principles and theories of management, organizational planning and strategy and acquiring, developing and retaining a skilled and diverse workforce.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Students will understand the core concepts, processes, principles, and significance of management and administration, and will be able to differentiate between the roles of managers at various levels within an organization. CO2. Students will be able to trace and critically evaluate the evolution of management thought, from classical theories to modern management concepts. CO3. Students will master the art and science of planning and decision-making in management, learning to establish objectives through Management by Objectives (MBO) and apply rationality in decision-making processes. CO4. Students will gain the ability to design effective organizational structures, understand departmentation, and manage staffing processes including manpower planning, recruitment, selection, placement, and induction. CO5. Students will explore various theories of leadership and motivation, develop skills to effectively direct teams, and engage with contemporary issues in leadership, including transformational, transactional, charismatic, and visionary leadership. CO6. Students will be equipped to manage strategic change, embrace knowledge management, foster learning organizations, and appreciate the value of managing diversity within the workplace.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies, etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii). Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT – I

Introduction: Concept of Management, Process, Principles, Levels, Functions and Significance of Management, Management Vs. Administration, Role of Managers. Evolution of Management Thought: Classical, Neo- Classical Theory, Behavioural Sciences, Approach, Quantitative, Systems and Contingency Approach, Modern Management Thought (Likert, Drucker, Porter, Prahalad).

Planning: Concept, Process and Significance, Types, Relationship between Planning and Controlling. Decision Making; Concept, Types and Process, Effective Decision, Rationality in Decision Making, MBO.

UNIT – II

Organization: Concept, Process and Significance, Principles, Organization Design and Classical Theory, Departmentation, Bases, Span of Control (Classical Theory and Situational Approach), Delegation of Authority, Principles, Centralization and Decentralization, Line and Staff Organization.

Staffing: Concept, Manpower Planning, Recruitment; Concept and Sources Selection; Concept, Selection Process and Tests, Placement and Induction.

UNIT – III

Direction and Motivation: Concept, Principles, Effective Supervision, Techniques. Motivation: Content Theories of Motivation, Current Issues in Motivation.

Leadership: Concept, Difference between Leadership & Management, Theories of Leadership (Trait Theory, Behavioural Theory, Managerial Grid), Contemporary Issues in Leadership (Transformational, Transaction, Charismatic and Visionary Leadership).

UNIT – IV

Coordination: Concepts, Importance, Internal – External Coordination Control: Concept, Steps, Types of Controlling, Techniques of Controlling. Management in Perspective (A Brief Overview), Management of Strategic Change, Knowledge Management, Learning Organization, Managing Diversity.

12. Suggested Resources for Reading

1. Drucker, P.F., The Practice of Management
2. Weihrich, H. and Koontz, H., Essentials of Management: An International & Leadership Perspective
3. Stoner, A.F. and Freeman, R.E., Management
4. Hampton, D.R., Modern Management
5. Robbins,S.P. & DeCenzo, D.A., Fundamentals of Management- Essential Concepts and Applications
6. Rao, VSP& Hari Krishna, V., Management Text & Cases

Note: Latest edition of text books may be used.

	COURSE TITLE	COMPUTERISED ACCOUNTING AND ENTERPRISE RESOURCE PLANNING
1.	**Course Code**	BLM 103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5 .	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Logistics Assistant
7.	**Course Objectives**	The objective of this paper is to provide students with a comprehensive understanding of computerized accounting systems including the benefits and limitation of computerized accounting, role of ERP systems in modern organizations and features and functionalities of ERP systems.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Students will be able to explain the meaning, importance, and differences between computerized and traditional accounting, as well as understand the underlying principles and limitations of computerized systems. CO2. Students will gain the skills to accurately record financial transactions using computerized systems, including the preparation of journals, ledgers, trial balances, cash books, and bank reconciliation statements. CO3. Students will understand the concepts and methods of accounting for depreciation, and be able to prepare final accounts, including trading, profit and loss accounts, and balance sheets for sole proprietorships, using accounting database management tools. CO4. Students will gain comprehensive knowledge of ERP systems, including their functionalities, benefits, typical business process workflows, and their impact on modern business concepts such as e-commerce, business analytics, and service-oriented architecture. CO5. Students will learn to navigate the challenges and considerations involved in ERP implementation, including system purchasing or outsourcing decisions, managing implementation partners, and understanding ERP strategy options and risk mitigation.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies, etc

10. Instructions for paper setters and candidates

i) The syllabus of this paper has been divided into FOUR units.
ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
iv) All questions carry equal marks.

11. Course Content

UNIT – I

Introduction of Computerised Accounting: Basic of Computerized Accounting: Meaning Importance Difference of Computerized and traditional accounting; Origin, Meaning, Definition, Need, Importance, Functions, Limitations, Accounting principles, Generally accepted accounting principles, Accounting equation, Double entry system.

Recording Transactions: Journal, Ledger and Trial Balance

UNIT – II

Cash Book and Bank Reconciliation Statement.

Accounting for Depreciation: Meaning, Importance, Methods of providing depreciation (straight line, diminishing, annuity), Reserves and Provisions

Final Accounts: Trading, Profit and Loss Account and Balance Sheet of a Sole Proprietary Concern. Accounting Database Management MS excel and its applications in Accounts and Statistics. Brief Introduction of different software available Tally 6.3 Busy Wings SAP etc.

UNIT – III

Introduction to ERP: Overview, Typical Business Process Workflow in ERP, Fundamentals, Functionalities, Benefits, Modern Business Concepts, E-Commerce, Advance Planning and Scheduling, Business Analytics, Service oriented architecture.

UNIT – IV

Issues, Concerns and Purchasing: Disadvantages of ERP Solutions, Users, Developers, Customers of ERP, Purchasing or Outsourcing, Planning, Purchasing and Selection of ERP, Managing implementation partners, ERP strategy options and risk mitigation.

12. Suggested Resources for Reading

1. Anthony, R.N., and J.S. Reece, Accounting Principles, Richard D. Irwin, Inc.
2. Monga, J.R., Financial Accounting: Concepts and Applications, Mayoor Paper Backs
3. Shukla, M.C., Grewal, T.S. and Gupta, S.C., Advanced Accounts, Vol-I, S. Chand & Co.
4. Gupta, R.L. and Radhaswamy, M., Advanced Accountancy, Vol-I, Sultan Chand & Sons,
5. Maheshwari, S.N. and. Maheshwari, S. K., Financial Accounting, Vikas Publishing House
6. Sehgal, A., and Sehgal, D., Advanced Accounting, Part -I, Taxmann Applied Services
7. Pandey, A., Enterprise Resource Planning, Variety Publishers.
8. Garg, V.K., and Venkita Krishnan, R.K., Enterprise Resource Planning: Concepts and Practice, Prentice Hall India.
9. Sumner, M., Enterprise Resource Planning, Pearson Education.
10. Leon, A., Enterprise Resource Planning, Tata McGraw Hill.

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	TRANSPORTATION AND DISTRIBUTION MANAGEMENT
1.	**Course Code**	BLM 201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv)Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided bythe University/College to the examinees
6.	**Job Role**	Logistics Assistant
7.	**Course Objectives**	The objective of the course is to help students understand the role and significance of transportation and distribution management, warehousing and distribution centre management, global transportation and logistics and apply concepts to real-world case studies.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Students will understand the role of distribution within the supply chain, including transportation and warehousing concepts, and be able to design distribution channels. CO2. Students will be adept at analyzing various factors affecting distribution networks, optimizing distribution networks using different approaches and techniques, and understanding delivery lead times and local facilities. CO3. Students will gain knowledge of transportation principles, the contributions of various transportation agencies, and manage transportation's role in material management processes. CO4. Students will be able to evaluate different transportation models, assess their performance characteristics, and understand the cost components of multimodal transportation. CO5. Students will be skilled in making transportation routing decisions, administering transportation operations, and applying information technology in transportation, including E-commerce and intelligent transport management systems. CO6. Students will be proficient in using transit operation software, understanding geographic information systems, advanced fleet management systems, intermodal freight technology, transport security initiatives, and the role of technology in transport inspections.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies, etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

iv) All questions carry equal marks.

11. Course Content

UNIT – I

Introduction: Role of distribution in supply chain – transportation management – warehousing concepts – designing distribution channels – understanding distribution costs, Advantages of distribution models – disadvantages of distribution models – pre-requisites of distribution – comparison of distribution networks.

UNIT – II

Distribution Network Planning: Various factors in distribution – delivery lead time and local facilities - optimization approach and techniques – material management process – role of transportation – transportation principles and participants – contribution of various agencies in transportation.

UNIT – III

Transportation Models: Performance characteristics and selection – various models of transportation (multimodal) – merits of each all models of transportation – transportation performance costs and value measures – understanding – comparing – cost components of multimodal transportation.

UNIT – IV

Transportation Routing Decisions: Transportation administration – transportation operations management – consolidation of freight – cost negotiations – various trends in transportation – application of information technology in transportation – E commerce – intelligent transport management system. Transit Operation Softwares: Geographic information systems – advanced fleet management systems- intermodal freight technology – transport security initiatives and role of technology – various inspection systems.

12. Suggested Resources for Reading

1. Rushton, A., Croucher, P., & Baker, P., The Handbook of Logistics and Distribution Management. Kogan Page.
2. Waters, D., Global Logistics: New Directions in Supply Chain Management. Kogan Page.
3. Hugos, M.H., Essentials of Supply Chain Management. Wiley.
4. Chopra, S., Supply Chain Management, Pearson.
5. Agarwal, D.K., Textbook of Logistics and Supply Chain Management, MacMillan Publishers.

Note: Latest edition of text books may be used.

	COURSE TITLE	INVENTORY MANAGEMENT
1.	**Course Code**	BLM 202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Logistics Assistant
7.	**Course Objectives**	The objective of the course is to help the students understand the importance and purpose of inventory management, various type of inventory, including raw materials, work-in progress, finished goods, spare parts, inventory cost management strategies and inventory optimization techniques.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Students will understand the importance, purpose, and scope of inventory control, and will be able to categorize different types of inventories, such as raw materials, work- in-progress, and finished goods. CO2. Students will learn strategies for managing various costs associated with inventory, including how to set up an organizational structure for effective inventory management. CO3. Students will gain proficiency in selective inventory control techniques, comprehend economic order quantity (EOQ), safety stocks, and apply forecasting techniques for materials requirement planning. CO4. Students will understand the principles of Manufacturing Resource Planning (MRP-II), Just in Time (JIT) inventory systems, make-or-buy decisions, and the role of outsourcing in inventory management. CO5. Students will learn various inventory optimization techniques and how to manage finished goods inventories, including the use of computers in inventory management and methodologies for performance evaluation of materials functions. CO6. Students will develop analytical skills through ratio analysis on inventory and profit margins, enhancing their decision-making capabilities in the context of core competency and inventory management trends.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies, etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT – I

Inventory: Inventory Management – Inventory Control – Importance and Scope of Inventory Control – Types of Inventory – Costs Associated with Inventory – Organizational set up for Inventory Management.

UNIT – II

Selective Inventory Control: Economic Order Quantity – Safety Stocks – Inventory Management Systems – Forecasting Techniques – Material Requirement Planning and Execution – Ratio Analysis on Inventory – Profit Margin.

UNIT – III

Manufacturing Planning (MRP-II): Just in Time (JIT) – Work in Process Inventories – Make or Buy Decisions – Concept of Outsourcing – Factors Influencing Make or Buy Decisions – Trends in Make or Buy Decisions in context of core competency.

UNIT – IV

Purpose of Inventory: Goods – Types of Goods – Finished Goods Inventories – General Management of Inventory – Stocks _ Types of Stocks – Tracking the Paper Life. Spare Parts Inventories: Use of Computers in Inventory Management – Evaluation of Performance of Materials Function – Criteria and methodology of evaluation.

12. Suggested Resources for Reading

1. Chopra, S., and Meindl, P., Supply Chain Management: Strategy, Planning, and Operation, Pearson.
2. Kumar, S.A, and Suresh, N., Inventory Management: Principles, Concepts and Techniques, Himalaya Publishing House.
3. Levi, D.S., Kaminsky, P., and Levi, E.S., Designing and Managing the Supply Chain: Concepts, Strategies, and Case Studies, McGraw-Hill Education.
4. Bowersox, D., J., Closs, D.J., and Cooper, M.B., Supply Chain Logistics Management, McGraw Hill Education.
5. Narasimha Murthy, M. L., Materials Management: An Integrated Systems Approach, Prentice-Hall of India.

Note: Latest edition of text books may be used.

	COURSE TITLE	WAREHOUSE MANAGEMENT
1.	**Course Code**	BLM 203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
4.	**Credits**	4 Credits
5 .	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Logistics Assistant
7.	**Course Objectives**	The objective of the course is to help students understand the concepts of warehousing in the supply chain, inventory management in warehousing, warehousing performance metrics and measurement, warehouse automation and technology and apply warehousing concepts to real-world scenarios.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Students will understand the key concepts, decision-making processes, operations, and the strategic role of warehousing in the supply chain. CO2. Students will gain a comprehensive understanding of the functions and types of inventory, the cost implications of holding inventory, and the competitive strategy of inventory management within the supply chain. CO3. Students will learn about the necessity and functionality of WMS, including logic for determining locations and sequences, and planning resources for distribution. CO4. Students will be able to apply ABC inventory control methods and understand the management of inventory within multi-echelon and single-echelon networks. CO5. Students will gain insights into the principles and performance measures of material handling systems, including automated storage and retrieval systems. CO6. Students will be adept in applying barcoding and RFID technology in warehousing scenarios, understanding their applications for improving warehouse efficiency.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Group Discussions, Case Studies, etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT – I

Introduction to Warehousing: Concepts – Decision making – Operations – Need for warehousing – Issues affecting warehousing – Various warehousing facilities – Different types of ware houses – Characteristics of ideal ware houses.

UNIT – II

Introduction to Inventory Management: Role in supply chain – Role in competitive strategy - Role of inventory – Functions of inventory - Types of inventory – WIP inventory – Finished goods inventory – MRO inventories – Cost of inventories - Need to hold inventory.

UNIT – III

Warehouse Management Systems: Introduction – The necessity of WMS – Logics of determining locations and sequences – Independent demand systems – Uncertainties in material management systems – Dependent demand systems – Distribution resource planning. ABC Inventory Control: Managing inventories by ABC – Multi – echelon inventory systems - Managing inventory in multi echelon networks – Managing inventory in single echelon networks. Various approaches – Distribution approaches – The true multi echelon approach.

UNIT – IV

The Principles and Performance Measures of Material Handling Systems: Introduction. Vehicle travel path(time) – Handling time – vehicle utilization – no of loads completed – congestion – Effective performance systems – Fundamentals of various types of material handling systems – automated storage and retrieval systems Bar coding technology and applications RFID technology.

12. Suggested Resources for Reading

1. Christapher, M., Logistics and Supply Chain Management. Pearson.
2. Raghuram, G. Logistics and Supply Chain Management. Mac Millan.
3. Richards, G. Warehouse management: A complete guide to improving efficiency and minimizing costs in the modern warehouse. Kogan Page.
4. Tompkins, J. A., & Smith, J. D. Warehouse management handbook. Tompkins Press.
5. Ackerman, K. B. Essentials of warehousing. Distribution Group.
6. Gue, K., & Meller, R. Material handling principles and practices. CRC Press.
7. Mulero, J. M. Facilities planning for warehouse operations. CRC Press.
8. Rowland, A., & Emmett, S. Modern warehouse management: A practical guide. Kogan Page.
9. Stephens, M. P. Material handling and facilities planning. Prentice Hall.

Note: Latest edition of text books may be used.